

$$f(x,y,z)=3x^2+2z-4y^2 \quad (3,-1,0) \quad \text{G, H}=?$$

$$G=\begin{bmatrix} 3x^2 \\ -4y^2 \\ 6xz+2-8yz \end{bmatrix}$$

$$H=\begin{bmatrix} 0 & 0 & 6z \\ 0 & 0 & -8z \\ 6z & -8z & 6x-8y \end{bmatrix} \quad H=\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 26 \end{bmatrix}$$